What do I want my message to be?
Chapter Overviews
- My Takeaways from each chapter
- My Unique experience – seeing a product from different angles and perspectives
- Working with different products and environments

BD Specific
- Moving with girlfriend
- Aligned Company Mission & Values
- Balance of hands on & design
- Transferable skills to the role
- Growth of working in a new environment & new materials & industry challenges

Anticipated Questions
- Industry Transition – My response to this would be that it wouldn't be the first time

Material & Testing Experience
- Laser Weld
    - Tensile Testing
    -
- Supplement
    - Vibration Testing
    - Thermal Cycle Testing

Next Steps:
- Laser Weld Overview Touch Up
- Story Arsenal
    - XRD
    - Capital Laser Weld Project
    - Goldwing V Story
- Resume Route
- Design of Experiment Walkthrough
- Flashcards Practice
- Review what Anita highlighted
- Behavioral/Situational/Technical

Tool belt:
- About Me
- Chapter Overviews
    - Mfg
    - Process
    - Tooling
- BD Specific
    - Company
    - Role
- Anticipated Questions
    - Industry Transition
- Material & Testing Experience
    - Laser Weld
        - Materials
        - Testing

    - Supplemental Mfg & Tooling
- Non Linear Material & Plastics
    - Personal Research
    - 3D Printing
- Story Arsenals
    - XRD Story
    - Capital Laser Weld Project
    - Goldwing V Story
- Behavioral Q's
- Situational Q's
- Technical Q's
    - Engineering Cheat Sheet
- Q's for them
- Adjacent Topics
    - Fatigue
    - Stress
    - Vibration & Thermal Cycling

Laser Weld Role:

Context:

- Process Engineer for Laser Weld
- Laser Welding is a portion of the build that seals the housing so the product can perform at cold operating temperatures

What:

- Process Monitoring / Statistical Process Control
- Dispositioning Defects and Root Cause Analysis with Program
- Develop Weld Recipes for new programs & optimizing welds
- Purchasing Capital Equipment
- Materials
    - Different types of Weld Compatibility
    - Surface Finish Testing – Material Composition (If I still have the email thread with Albert Linder)
    - Materials & Process Lab Testing Equipment (Adhesives, Shear Testing
- Testing
    - Coupon Testing – Health of our machine
    - Sample Testing – Bimonthly Program Recipe Verification
    - Product Testing – Helium Leak check
    - Strength Testing – Tensile Testing
    - Non Destructive Testing Case – XRD – X-Ray Diffraction for a customer
    - Rework: Cut-open Procedure

Takeaway:

- Laser Welding Process taught me a lot about diving deep into a process and understanding
    - KPIs – Key Performance Indicators
    - Cpk & SPC – Working with data

- Being responsible for a

Remaining Actions:

- Determine ISO Standards that were used and applied
- IPG Laser Welder Specifics (Laser Power / Fiber Optics / Argon Shielding)

https://amadaweldtech.com/wp-content/uploads/2023/04/Welding-Material-Control.pdf

Laser Weld

Fundamentals

Establish Terminology for Laser Welding Process using a photo

- Conduction vs Keyhole Welding
- Absorption & Reflectivity of Metals
- Power Density, Focal Spot Size, Energy Input
- Melt Pool Dynamics & Solidification
- HAZ – Heat Affected Zone

Industrial Laser Types

Different types of Lasers

- Fiber - ideal for high-speed, precise metal cutting/marking with low maintenance
- Nd-YAG - specialized, high-intensity lasers often used for deep metal engraving and precise welding
- CO2 - excel at cutting/engraving organic materials (wood, acrylic) and thick non-metals

Fiber Lasers: Use fiber optic cables with rare-earth elements (ytterbium), offering high energy efficiency (>30%) and virtually maintenance-free operation. Best for thin-to-medium metal processing

CO2 Lasers: Use a gas mixture (CO2, N2, He), offering high power for non-metal cutting and engraving. They require more maintenance due to mirrors and are generally less efficient than fiber, but handle thick, non-metallic materials better.

Nd:YAG Lasers: Use a neodymium-doped crystal (Yttrium Aluminum Garnet), ideal for metal marking and welding due to high beam intensity. They have higher maintenance and lower energy efficiency compared to fiber lasers.

Heavy industry and high-power applications

Comparison Table

Feature	Fiber Laser	CO2 Laser	Nd:YAG Laser
Wavelength	1.06 µm (Near-IR)	10.6 µm (Far-IR)	1.06 µm (Near-IR)
Primary Material	Metals (steel, copper)	Organics (wood, plastic)	Metals & some plastics
Beam Quality	Very High	Moderate (divergence)	High
Efficiency	Very High (> 30%)	Low (< 10%)	Low
Maintenance	Low/None	High (mirrors, gas)	Moderate/ High
Key Advantage	High speed, precision	Versatility, thick cutting	High intensity/ deep marking

Explain Why Fiber Lasers Dominate Modern Welding Lines (Beam Quality, Efficiency, Fiber Delivery)

Fiber Lasers have near perfect M2 factor – allows for smaller, higher intensity focal spot
M2 - measures how closely a beam approaches an ideal theoretical laser. A lower ensures a tighter focus, higher power density, and deeper, faster, more precise penetration

M2:
Imagine you are using a garden hose to water a plant.

- If you have a **nozzle** that makes a perfect, tiny, and powerful stream of water, you can hit one exact spot, and the pressure is very high.
- If you just have the **open hose**, the water sprays out, spreads quickly, and is not very concentrated.

In laser welding, the M2 **factor (M-Squared)** is a number that tells you how much your laser behaves like that perfect, tiny nozzle versus the messy spray.
The Core Idea (ELI5)

- **(The Perfect Beam):** This is the ideal laser. It can be focused down to the smallest possible, most intense, and hottest spot. This is great for deep, precise welding.
- **(The Real World):** No real laser is perfect. The higher the number (e.g., 2, 3, or 4), the more the laser spreads out and the less intense the spot becomes.
- **Lower is Better:** A lower M2 means a sharper, more focused, and more efficient welding beam.

Why does M2 matter for Welding?

If your M2 is low (close to 1):

1. **Deeper Welds:** The laser is concentrated, so it can melt deeper into the metal.
2. **Faster Welding:** You can move faster because the energy is packed tighter.
3. **Smaller Heat Zone:** You only melt exactly where you want to, so the surrounding metal doesn't get damaged (less warping).

Summary
Think of M2 as a **focus quality score**.

- **Low M2 (e.g., 1.1):** Laser Ninja (sharp, precise).
- **High M2 (e.g., 3.0):** Laser Squirt Gun (spread out, less precise).

Process Parameters & Control

- Laser Power
- Travel Speed
- Focal Position
- Shielding Gas Type & Flow
- Beam Oscillation & Wobble
- Joint Fit-up & Gap Tolerance

- Material properties: composition, reflectivity, coatings, and oxide layers
- Process parameters: peak power, pulse energy, pulse width, shielding gas, spot size, repetition rate, and weld speed
- Equipment: laser source, beam delivery, motion control, and part fixturing

Process Windows & Sensitivity

- Parameter Matrices
- Tradeoffs between penetration, spatter, porosity
- Statistical Control of Weld Parameters

Materials Knowledge

- Stainless Steels
- Nickel Alloys
- Aluminum Alloys
- Titanium Alloys

Materials with different melting points/expansions (steel to aluminum) will lead to cracks & distortion
Titanium is extremely oxygen-sensitive – need to tighten shielding to protect from contamination

Example Material Challenges

- Hot Cracking
- Porosity in Aluminum
- Reflectivity Issues in Copper & Aluminum
- Oxygen Sensitivity of Titanium

Weld Defects & Root Cause Analysis

- Weld Gaps
- Porosity – Trapped gas bubbles in welding pool caused by surface contamination
- Discoloration - caused by surface oxidation (chromium oxide) resulting from heat and insufficient shielding gas coverage
- Cracking – High Speed & High Intensity Heat can cause hot or cold cracking
- Spatter

https://amadaweldtech.com/wp-content/uploads/2025/10/Closing-the-gap-overcoming-common-challenges-in-laser-welding-medical-devices.pdf

Inspection Methods

- X-Ray Radiography
- Ultrasoinic Testing
- Dye Penetrant Inspection
- Computer Tomography

Destructive Testing (Sample-Based)

- **Cross-Sectioning/Acid Etching:** A sample weld is cut, polished, and etched to inspect penetration, fusion, and porosity.
- **Tensile/Bend Testing:** The weld is physically tested for strength, usually done on test samples rather than every production part.

Non-Destructive Testing (NDT) (Post-Process)

- **Visual Inspection:** The initial check for surface issues like cracks, undercut, or incomplete filling.
- **Radiographic Testing (RT):** Ideal for identifying internal, microscopic flaws and porosity, providing a permanent record of the weld quality.

https://www.youtube.com/watch%3Fv%3Dsd0t40NXQAY
https://www.youtube.com/watch%3Fv%3D7oR_l8nXdmo

Real-Time Processing

- **Camera-Assisted Vision Systems:** Cameras monitor the weld pool, bead consistency, and color, which is particularly useful for identifying gaps or improper, oxidized welds.

Aerospace Quality & Standards

- AWS D17.1
- AS9100

Process Qualification vs Production
First Article Inspection
Traceability
Weld Procedure Qualification Records

Process Development & Qualification (R&D to Production)

- Design DOE Experiments
- Defining Stable Process Window
- Define Control Plans
- Run Capability studies (Cp, Cpk)
- PFMEAs

Automation & Manufacturing Integration

- Vision Alignment
- Real-Time Monitoring (Photodiodes, Melt Pool Cameras)
- CNC Control

Safety

- Laser Classes
- Interlocks & Enclosures
- Fume Extraction

Why Argon?
Argon is considered the best welding gas because it is a cheap, dense, inert noble gas that provides superior arc stability, excellent shielding from atmospheric contamination, and minimal spatter, resulting in cleaner, high-quality welds

- Cheap
- Dense
- Inert Noble Gas = Stable & Nonflammable

Scenarios:

- Design a laser weld process for a thin titanium butt joint and walk through:
- Parameter Selection
- Shielding Strategy
- Fixturing
- Inspection Plan
- Failure Modes

Examples of Material Testing Experience Throughout Career

- [MFG] – Coupon Shear Testing
- [MFG] – Material Selection within Dewar Product
- [MFG] – Vibration & Thermal Cycle Testing
- [M&P] – Welding Material Compatibility
- [M&P] – Laser Welding Process Sustainment
- [M&P] – XRD Case Study
- [TE] – Material Selection within Tooling Design, Composites?
- [TE] – 3D Printing Materials
- [TE] – FEA, Stress, Fatigue Analysis

Design of Experiment Walkthrough

- **Define:** Identify the experimental purpose, response(s), and factor(s).
- **Model:** Propose or specify an initial statistical model.
- **Design:** Generate and evaluate an experimental design that can support the initial statistical model.
- **Conduct:** Execute runs and collect data for each row (run) in the design.
- **Analyze:** Fit a statistical model to the experimental data to answer your research questions.
- **Predict:** Use your confirmed statistical model to predict future values of the responses.

https://www.jmp.com/en/statistics-knowledge-portal/design-of-experiments/design-of-experiments-workflow

Resume Routing:

Tooling Design
- 3D Modeling
- Tolerance Stack Up
- DFM Principles
- Design Intent & Physical outcomes ➔ World of Modeling vs World of Reality
- FEA Ansys (static/dynamic/thermal loading conditions)
- Stress Analysis
- Fatigue Analysis
- PLM (Product Lifecycle Management)

Materials & Process
- Metallurgical Analysis
- Thermal Analysis
- Hardness Testing
- Microstructure Evaluation
- Custom Test Fixtures
- SPC & Cpk Analysis
- DOE-based characterization workflows
- Capital Equipment Project

Semiconductor
- Automated Assembly Lines
- Scanning Electron Microscopy (SEM) → Wirebond
- PRR/CDR/TRR Reviews
- V&V Frameworks

Manufacturing
- EVMS & Budgets
- Class 7/Grade C/ Lvl 3 Cleanroom
- Additive Manufacturing

Project Arsenals
- XRD Story
- Capital Laser Weld Project
- Goldwing V Story

Process Improvement
Leadership / Ownership

Capital Laser Weld Project

- Context:
    - Our site had two laser welding machines: One was a modern IPG Welder bought in 2015. And the other was built in the 1990's basically on life support and it was getting more issues with calibration and parts were becoming obsolete. The machine was a risk to the factory if it were to go down, it would create a single point failure.
- Key decision:
    - We decided to go with the same company as our existing welder to be consistent with our existing welder.
- My Role:
    - My role as the Process Engineer was to essentially lead the to purchase a Laser Welding machine for the site
    - This included providing justification for a new machine, working with the supplier for a new machine, coordinating with factory and facilities to schedule an installation plan, notifying the affected programs and creating productions, presenting to PCCB,
        - Testing: Running Coupons of existing programs, verifying any deviations
- Result:
    - It took a few months of rigorous work but we eventually got the new laser welder in. There were some minor hiccups along the way (fixtures, lens focal length, figure 8 feature)
- Lesson:
    - What I thought was going to be a relatively simple process of copy and paste turned to be one of the most extensive projects mainly because it was a capital equipment requires more scrutiny and justification for purchase & half a million-dollar equipment
- That's it

Cross-Conflict Problem & Failure & Recovery & Didn't know what to do
XRD Project

- Context:
    - Our site had a new contract for a program and the customer had really strict requirements. When it got to Laser Welding, we have our own mitigations of process control and specifications – and they gave us samples and our weld
        - Low Volume // Hard to get engineering samples
        - Wanted a new procedure on verifying the laser weld depth besides the Destructive Tests
        - Not willing to budge but Program Manager said this was an important contract

- Key decision:

- We explored many options:
    - Working with IPG (the laser weld supplier) if their software could measure the depth of a laser → They said yes but we would essentially need to get a new machine (which ironically we eventually did)
    - Working with the customer to supply us smaller scaled housings for us to destructively test → Customer didn't think the smaller scale would be truly represented due to heat dispersion
- Key Decision we all agreed on was a XRD test where

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Story Telling:

- Context:
- My Role
- Key decision
- Result
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BD Preparation AM

Laser Weld Role:

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Design of Experiment Walkthrough Template

- Define Objective
- Select Factors
- Brainstorm & Choose Levels
- Design & Run Experiements
- Analyze Data
- Optimize and Validate

Resume Routing:
Tooling Design
- 3D Modeling
- Tolerance Stack Up
- DFM Principles
- Design Intent & Physical outcomes ➔ World of Modeling vs World of Reality
- FEA Ansys (static/dynamic/thermal loading conditions)
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Story Telling:

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- That's it

Plastics Video

-mer (unit) → Polymer

Atom Structure of different Polymers: Carbon, Hydrogen, + Fluorine/

Two Polymers with same chemical structure could have two different

Polymerization – Process that turns monomers into polymers

Different plastics use different resources

PLA: lactic acid

Copolymer – using different monomers

ABS – A = , B = , S =

Two types of Polymerization: Addition & Condensation

Addition: Chain Growth Process

Condensation: Two different types of monomers

Step Growth Process

Linear Arrangement – Weak

HDPE, PVC

Branched Arrangement –

LDPE

Cross-Linked Arrangement – Rigid 3D Network

Vulcanized Rubber

Thermoplastics (Polymers that soften or melt when heated)
Widely used
Can be heated and reshaped for injection molding
Can't perform at high temperatures

Thermosets
Cross-Linked
Do not melt
Used in polymer fibers in composites
Legos

Crystallinity
Unlike Metals, Plastics are Semi-crystalline
Regions of Amorphous Regions & Crystalline Patterns
Layers of Order & Jumbled

Linear = High Crystallinity
HDPE, PVC
Stronger, Stiffer, Higher Melting Point
Branched = Low Crystallinity
LDPE, PEEK
Cross-Linked = No Crystallinity
PMMA (Plexiglass)
Will be transparent because Crystallinity scatters light so amorphous is more transparent

*Crystallinity explains different material properties despite same chemical structure of repeating mer unit

Ashby Chart of Stiffness & Density:
Polymers:
Advantage:
Lightweight
Thermal & Electrical Insulation
Corrosion Resistance
Low Cost
Disadvantage: Low Strength & Stiffness

Elastomers:
Latex – Natural Rubber
Weak Interconnections with polymers allow it to be stretchy and deform

Vulcanization: Can improve performance

Heating Rubber with sulfur which forms cross-links between polymer chains

Creates anchor points which increases strengths

But tricky balance because if too many anchor points will become like a thermoset and brittle and not be able to stretch as desired

Vulcanization common in Elastomers like Gaskets & Tires to enhance strength & durability

Stress Strain Curve Analysis

Brittle: (Epoxy Resins, PMMA, PLA)

Ductile: (PE, PP, Nylon)

Elastomers: Different Curve

Ductile Polymers

Necking isn't like Metals

Experiences Cold Drawing at Necking which momentarily aligns fibers/polymer chains and actually strengthens the material

Kevlar – Uses Cold Drawing Behavior to intentionally drawn to create material with high tensile strength along the fiber axis

Brittle Polymers

Crazing - tiny voids form within the polymer just before fracture with fibrils

Unlike cracks in metal

Crazing can be an indicator of catastrophic failures soon to happen (unlike metals)

Material Properties with Temperatures

Low Temperature: Stiffness is high & stable

Glass Transition Temperature: hard to soft transition temperature

Increase Temperature: Stiffness decreases → Soft Compliant Material

Melting Temperature

Liquid: No Stiffness

Amorphous Polymers – no well-defined melting points

Thermosets – don't have melting point at all

Glass Transition Temperature Factors: Anything that restricts polymer chain mobility will increase T_g

Cross-Linking

Bulky Side Groups

Long Entangled Polymer Chains

Plasticizers – additive small molecules that insert themselves between polymer chains to reduce intermolecular forces to reduce Glass Transition Temperature
Plasticizing PVC to be flexible at room temperature used for wire insulation and flexible tubing

Case Study: Challenger Space Shuttle O-rings at colder temperatures became more stiff and improper seal

Viscoelasticity – Behaviour where material behaves partly

Laser Weld Graphic
- Top: The What & Why & The
  - Referenced Resources ASTM or ISO
- Sections:
  - Testing (See Claude)
    - Coupons
    - Sample Parts
    - XRD

Material
- Weld Compatibility Matrix
- Welding General Rules & Best Practices
  - (Bright

<u>The Power of Gold</u>
https://www.garfieldrefining.com/resources/blog/how-is-gold-used-in-space-exploration/
Spacesuits
NASA also uses gold in the construction of spacesuit visors. A very thin layer of gold is applied to the outside of a polycarbonate plastic visor to protect astronauts from the sun's harmful radiation in space.
Gold is ideal because it reflects nearly all infrared light, which can damage the eyes – even though we can't see or feel it. By itself, gold transmits about 60% of visible and ultraviolet (UV) light, so it doesn't provide full protection.

That’s where polycarbonate plastic comes in. It blocks most UV light, which can also damage the eyes. Together, the gold coating and polycarbonate visor block harmful infrared and UV rays while still allowing enough visible light through for astronauts to see!

Technical win	Describe a challenging engineering problem you worked on.	**XRD Test / BL0 Project**
Failure or recovery	Tell me about a time you failed or made a mistake. What did you learn?	**XRD Test / MRSI Component**
Cross-functional conflict	Describe a time you worked on a team with conflicting opinions.	**XRD Test / Goldwing**
Tradeoff decision		**BL0 Project**
Process Improvement		**Capital Laser Weld**
Leadership / Ownership		**Capital Laser Weld**

Behavioral

Tell me about yourself.	[Insert Elevator Pitch]
Tell me about a time you failed or made a mistake. What did you learn?	Right is better than being rushed Delegating is better than trying to do it all Done is better than perfect
How do you handle tight deadlines or multiple projects?	
Give an example of when you had to explain a technical concept to a non-technical audience.	
Tell me about a time you took initiative or led a project.	
How do you handle feedback or criticism?	Understanding the bigger picture like the Why Giving a timeline or quantifiable metrics to measure past / future performance

Situational Questions (logic, ethics, communication, and practicality) "what would you do if…" questions to test judgment and decision-making.

What would you do if your design didn't meet safety	Acknowledge → Assess → Act → Communicate → Prevent Recurrence

requirements late in the project?	What they're testing: Ethics, safety mindset, courage under pressure. I'd assess the gap, propose corrective design changes, and evaluate the schedule and cost impact. I'd work with stakeholders to find the safest path forward rather than pushing something that could fail in the field. Example: Safety Project for the PCUs
How would you handle discovering an error in your calculations after submission?	**Integrity & Accountability** "I'd verify the error first, understand its impact, and then notify the appropriate people as soon as possible. I'd explain the mistake, provide corrected calculations, and outline whether it affects the design, safety, or timeline. I believe transparency builds trust, even when it's uncomfortable.
If a teammate isn't pulling their weight, how would you handle it?	**Team Skills & Conflict Management** "I'd start by having a private, respectful conversation to understand if there's a workload, skill, or communication issue. If needed, I'd offer help or clarify expectations. If the issue continued and affected the project, I'd bring it to the team lead with a focus on solutions, not blame." Example: Working with V on Goldwing
How would you prioritize tasks when multiple projects are due at the same time?	**Time Management & Judgement** "I'd prioritize based on deadlines, technical risk, and impact. I'd break tasks into smaller deliverables, identify critical paths, and communicate early if conflicts exist. If necessary, I'd align priorities with my manager so expectations are clear." Example: Tooling Projects // Process
What would you do if manufacturing tells you your design is too expensive to produce?	**DFM Mindset & Collaboration** I'd work with manufacturing to understand the cost drivers—materials, tolerances, processes—and look for design alternatives that meet performance requirements at lower cost. I see manufacturing feedback as part of optimizing the design, not a roadblock. Propose different solutions with the understanding of different risks
How would you respond if your manager disagreed	**Professionalism & Influence without ego** "I'd make sure I clearly explain my reasoning, data, and assumptions. If my manager still disagreed, I'd listen to their

with your technical recommendation?	perspective and understand their constraints. Ultimately, I'd support the final decision while documenting risks if appropriate." Clearly document the risks

Technical Questions

Bonus "Sneaky" Questions (These blend everything together):

Walk me through one of your projects from concept to completion.	
What's your favorite engineering project and why?	
What would you improve in this design? *(they may show you a part)*	
How do you ensure safety and compliance in your designs?	**Asking for feedback from different perspectives (Bridging the gap between Engineering & Technicians)**
Process Overview	FPA Laser Weld Empennage

Analysis & Tools

Have you used FEA or CFD? Explain your process.	

How do you validate simulation results?	
Experience with MATLAB, Python, or Excel?	**MATLAB – Hexapod System** **Python – UW College** **Excel – Data Export**
How do you approach troubleshooting a mechanical system?	

Materials

How do you select materials for a specific application?	
Difference between steel, aluminum, and cast iron?	
What is heat treatment?	
What is creep?	
What are composites and where are they used?	

Manufacturing & Design

Difference between machining, casting, forging, and additive manufacturing.	
What are GD&T symbols and why are they used?	
Explain tolerance and fit (clearance,	

interference, transition).	
What CAD tools have you used? (SolidWorks, CATIA, Inventor, etc.)	
How do you design for manufacturability (DFM)?	

Mechanical Fundamentals

What is Young's modulus?	
Difference between ductile and brittle materials?	
What causes fatigue failure?	
What is factor of safety and why is it important?	
Explain torsion in shafts.	

Thermo / Fluids (if relevant)

Difference between conduction, convection, and radiation.	
What is Reynolds number?	
Explain the first law of thermodynamics.	
Difference between laminar and turbulent flow?	
What is efficiency vs. effectiveness?	

Company Specific

What excites me about [Alten Technology]	
What makes me a good fit for [Alten Technology]	
Questions I have for [Alten]:	- Work Life Balance - Relationship with Clients